

Data Mining Numerical Problems & Solutions

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · Chapter 7: Advanced Applications

Ch 7 – Advanced Applications – Numerical Problems & Solutions

8 numerical marks, 0.4% of the archive, 6% of this chapter · This chapter is almost entirely short notes. Across all 23 papers there is exactly one numerical 76 Ch, on PageRank. It is solved in full below, plus 1 practice problem on the variant most likely to be set next.

Both end with the solution drawn step by step. Each panel carries a chip naming the move it makes; the decisive step is framed **KEY STEP**.

The moves the chapter-7 numerical below is built from

Move	What it says	Where it is used (problem, step; orange = the key step)
outlinks vs inlinks	Outlinks give the divisor C ; inlinks say which terms to add.	1.1 1
split a rank	A page's rank is shared equally among the pages it links to.	1.1 1,2
use the old values	Compute every page from the PREVIOUS iteration, then substitute together.	1.1 3
no inlinks	A page nothing points at stays at $1 - d$ for ever.	1.1 2
order before value	The ranking settles long before the numbers stop moving.	1.1 4
sum check	In this form the converged ranks add up to N .	1.1 5
normalised form	$\frac{1-d}{N}$ instead of $1 - d$: the same ordering, ranks summing to 1.	Practice 1

M1. PageRank

Method

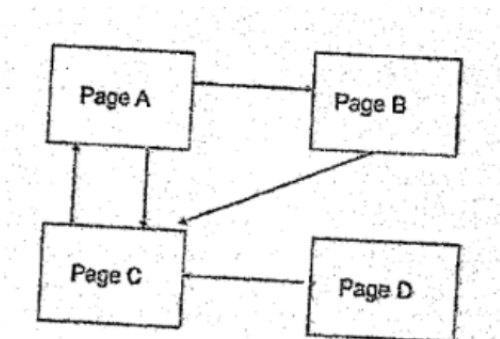
$$PR(A) = (1 - d) + d \sum_i \frac{PR(T_i)}{C(T_i)}$$

- Write down the link table first:** for every page, its **outlinks** (to get C) and its **inlinks** (to know which terms to sum). **Nearly every error comes from confusing the two.**
- Initialise all ranks (here 0.25).
- Each iteration, compute all pages using the **previous** iteration's values, then substitute in together.
- Repeat for the required number of iterations, or until the values settle.
 - A page's own rank is split equally among its outlinks** – that is the $/C(T_i)$.
 - A page with **no inlinks** sits permanently at $(1 - d) = 0.15$.

Problem 1.1 8

QUESTION AS PRINTED 2076 Chaitra · Q12 · [8]

Consider the following subset of pages and their links. Apply the PageRank algorithm using a damping factor of 0.85. A minimum of five iterations are required. Assume initial page rank of all pages is 0.25.



Step 1: read the graph

Page	Outlinks	C	Inlinks
A	B, C	2	C
B	C	1	A
C	A	1	A, B, D
D	C	1	none

Step 2: write the 4 equations

($d = 0.85$, $1 - d = 0.15$)

- $PR(A) = 0.15 + 0.85 PR(C)$
- $PR(B) = 0.15 + 0.85 \frac{PR(A)}{2}$
- $PR(C) = 0.15 + 0.85 \left(\frac{PR(A)}{2} + PR(B) + PR(D) \right)$
- $PR(D) = 0.15 + 0.85(0) = \mathbf{0.15}$ **constant, D has no inlinks**
- Note A's rank is divided by 2 because A links to *both* B and C.

Final answer after 5 iterations

- $PR(A) = 0.972$, $PR(B) = 0.519$, $PR(C) = 1.028$, $PR(D) = 0.150$
- **Ranking: C > A > B > D**

Step 3: iterate

Iter	PR(A)	PR(B)	PR(C)	PR(D)
0	0.2500	0.2500	0.2500	0.2500
1	0.3625	0.2563	0.6813	0.1500
2	0.7291	0.3041	0.6494	0.1500
3	0.7020	0.4599	0.8459	0.1500
4	0.8690	0.4484	0.9668	0.1500
5	0.9718	0.5193	1.0280	0.1500

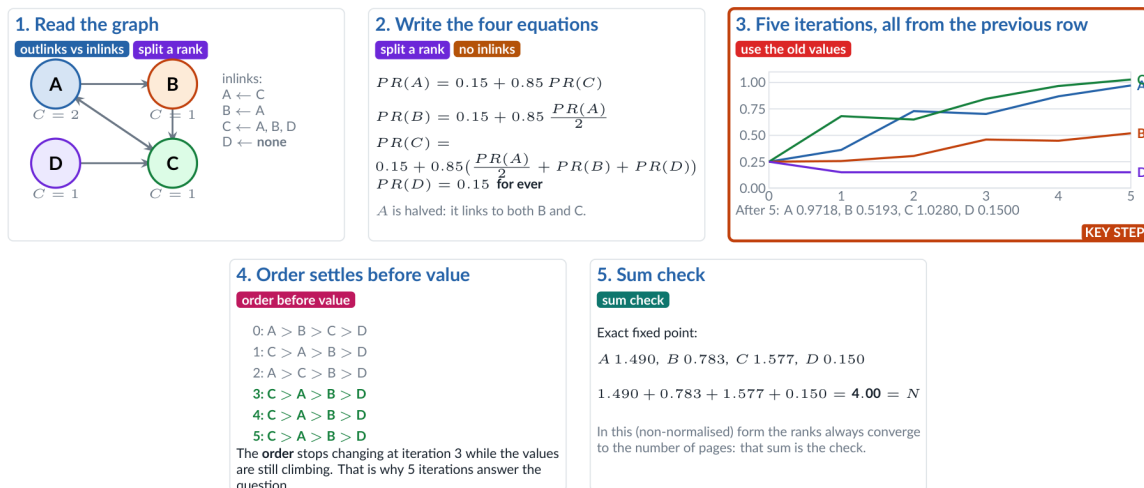
Row 0 is the given start value.

Sample working (iteration 1)

- $PR(A) = 0.15 + 0.85(0.25) = 0.3625$
- $PR(B) = 0.15 + 0.85(0.25/2) = 0.2563$
- $PR(C) = 0.15 + 0.85(0.125 + 0.25 + 0.25) = 0.6813$
- $PR(D) = 0.15$

Two comments worth writing, they show understanding. verified/added

1. The values are still climbing at iteration 5 but the **ranking has been stable since iteration 3**. PageRank converges on *order* much faster than on *value*, which is why 5 iterations is enough for the question.
2. Solving the equations exactly gives the fixed point $PR(A) = 1.490$, $PR(B) = 0.783$, $PR(C) = 1.577$, $PR(D) = 0.150$, and these sum to $4.00 = N$. **That sum is your check** – in this (non-normalised) form the ranks always converge to the number of pages.
3. **C wins** because it collects links from three pages while A collects only one, and **D is stuck at 0.15** because nothing points to it. Linking out never raises your own rank.

Step by step: the graph, the equations, five iterations, and the two checks**Practice – the normalised form**

PROBLEM not a PYQ · the same graph, normalised form verified/added

Repeat Problem 1.1 on the same four-page graph, but using the *normalised* PageRank formula

$$PR(A) = \frac{1-d}{N} + d \sum_i \frac{PR(T_i)}{C(T_i)}$$

with $d = 0.85$, $N = 4$ and an initial rank of 0.25 for every page. Compare the converged ranks and the ordering with those of Problem 1.1.

This form makes the ranks sum to 1 instead of N . Know both. verified/added

- With $N = 4$, $\frac{1-d}{N} = \frac{0.15}{4} = 0.0375$.
- Iteration 1 from the same 0.25 start:
 - $PR(A) = 0.0375 + 0.85(0.25) = 0.2500$
 - $PR(B) = 0.0375 + 0.85(0.125) = 0.1438$
 - $PR(C) = 0.0375 + 0.85(0.625) = 0.5688$
 - $PR(D) = 0.0375$
- Sum = 1.0001 ≈ 1 ✓ — **that is the check for this form.**
- **The ranking is identical** ($C > A > B > D$); only the scale differs.
- **In the exam:** state which form you are using in one line. Either is accepted; mixing them is not.

Step by step: the same graph in the normalised form (ranks sum to 1)

1. The normalised form

normalised form

$$\frac{1-d}{N} = \frac{0.15}{4} = 0.0375 \text{ replaces } 0.15$$

Iteration 1: A 0.2500, B 0.1438, C 0.5688, D 0.0375

Sum = 1.0001 ≈ 1 : the check for this form.

Same ordering $C > A > B > D$, only the scale differs.
State which form you are using.

KEY STEP

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity-relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	k -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /23** = topic asked in 7 or more of the 17 papers · **HOT /23** = 4–6 papers · **PIN /23** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. verified/added marks anything researched and added that was not in the source notes.